



Using control theory to study visuomotor control abilities in normal adults, athletes, and patients with neuromotor disorders

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Control theory, a field rooted in engineering and mathematics, aims to understand and influence the behavior of dynamic systems to achieve desired outcomes. Its principles are particularly suited for studying human visuomotor control, as humans inherently operate as dynamic systems. Visuomotor behaviors require the continuous integration of visual input with motor responses, often in the presence of disturbances or unpredictable changes. By applying control theory, researchers gain significant insights into how the visual system processes information to produce adaptive motor outputs, enabling the simulation and prediction of precise movements, such as eye and manual tracking, driving, and playing sports. This review highlights recent progress in applying classical control theory to investigate visuomotor control across diverse populations, including normal adults, athletes, and individuals with neuromotor disorders like Parkinson's disease. Future directions for research are also discussed, emphasizing the potential of control theory methods to deepen our understanding of visuomotor control and improve therapeutic interventions.

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Introduction

Control theory, a branch of engineering and mathematics, examines how systems — biological, mechanical, or computational — regulate their behavior to maintain stability and achieve specific goals in the presence of disturbances or noise [11,12,4,41]. At its core, control theory emphasizes closed-loop feedback, wherein continuous inputs and outputs interact dynamically. Models describing these relationships are often derived experimentally, enabling predictions about system behavior under various conditions (Table 1).

Human visuomotor control provides an ideal context for applying control theory, as humans inherently operate as dynamic systems. In natural environments, visuomotor behaviors demand continuous integration of visual input and motor output to respond to disturbances or unexpected changes. By combining experimental data, mathematical modeling, and computational analysis, control theory offers a structured approach to simulating and understanding these complex processes. Despite its potential, relatively few studies have rigorously applied control theory to test hypotheses in visuomotor research.

This review highlights widely used control theory methods and recent advances in studying visuomotor control across diverse populations, including normal adults, athletes, and individuals with neuromotor disorders such as Parkinson's disease (PD). Focusing on target-tracking tasks, I discuss the dynamics of human-in-the-loop systems, emphasizing compensatory control models. Furthermore, I explore the interplay between control theory and continuous psychophysics, which provides insights into the integration of perception and action.

The findings discussed here address fundamental questions about how humans integrate visual information to produce precise motor outputs. They advance our understanding of the underlying mechanisms of visuomotor control and inform practical applications such as rehabilitation technologies, robotic systems, and virtual reality interfaces. These tools hold promise for improving visuomotor function in clinical and everyday settings.

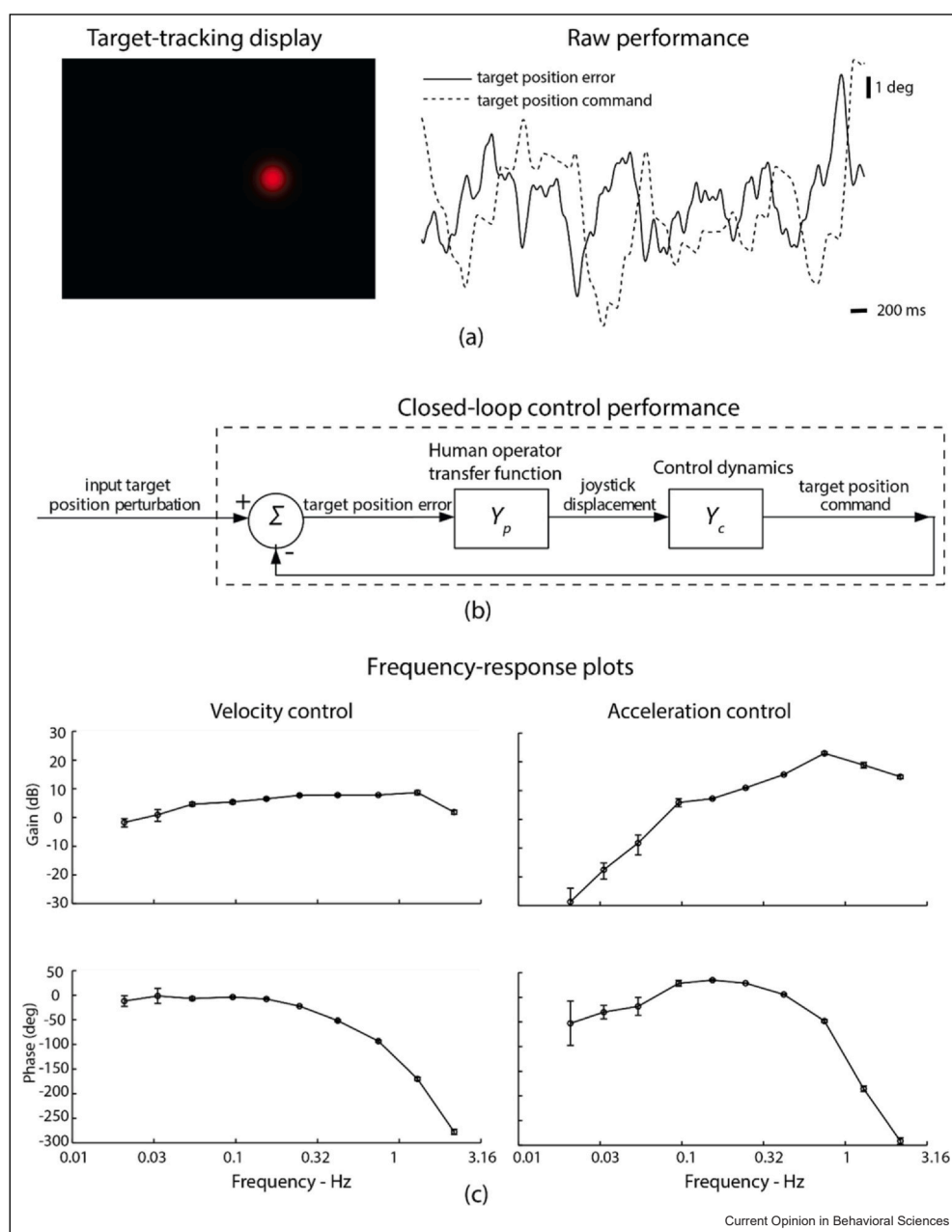
Background

Control theory provides a systematic framework to examine how varying visual inputs influences human motor control performance. A commonly used approach

involves closed-loop manual control tasks where participants (i.e. human operators) use a control device (such as a joystick or a mouse, also known as the plant) to

maintain a target position, perturbed pseudo-randomly by a sum of sinusoidal functions, near a pre-specified position (such as the center of a screen, see [Figure 1a](#)).

Figure 1



(a) An illustration of the display used in a target-tracking task. The target moves pseudo-randomly along the horizontal axis of the screen (i.e. its horizontal position is perturbed by a sum of sinusoidal functions). The target displays a rightward error from the center of the screen in the illustration (left panel). Typical participant raw performance data from part of a trial (right panel). The solid line depicts the input target position error, and the dotted line depicts the output target position command. Note that the output is a low-pass-filtered and delayed version of the input. **(b)** A simplified block diagram of the closed-loop target-tracking task. The system is modeled as a feedback loop in which the human operator responds to a target position perturbation (input) by generating a joystick displacement (output). Human operator transfer function Y_p captures the participant's control behavior, and the control dynamics Y_c specify how the displacement of the joystick generates commands over time. **(c)** Sample frequency-response plots of response gain and phase as a function of input perturbation frequency for velocity control (left panels) and acceleration control dynamics (right panels).

(a) Right panel, adapted from Ref. [31]. **(b)** Adapted from Ref. [26].

Despite its simplicity, this task requires the sensorimotor system to integrate visual information (e.g. target position and velocity) with predictions about the sensory consequences of the control device's displacement. Any failure in these steps leads to deteriorated control performance [25,44,48]. This task enables researchers to not only examine the specific contributions of visual cues (e.g. luminance and color contrast, depth cues, position, motion, etc.) but also assess the influence of training and neuromotor disorders on control performance metrics like precision, response gain, and phase lag.

Control dynamics describe how the displacement of a control device generates commands over time (Y_c , Figure 1b). In zero-order dynamics, displacement directly determines the target's position (i.e. position control), as in moving a mouse to change the location of a cursor. In first-order dynamics, displacement affects the target's velocity (i.e. velocity control), such as turning a steering wheel to change a car's heading. In second-order dynamics, displacement influences the target's acceleration (i.e. acceleration control), as in manipulating a control stick to adjust the pitch of a hovering aircraft [21]. Acceleration control is inherently more difficult than position or velocity control, as it involves higher-order derivatives that amplify system noise and error, increasing the challenge of maintaining stability and precision.

Previous studies have shown that human control behavior changes as the control dynamics varies to maintain overall system stability. For velocity control, humans typically act as gain controllers with time-delayed outputs (Figure 1c, left panels). In contrast, for acceleration control, although more challenging, through training, humans learn to behave more like differentiators that use rates of change to respond before errors fully develop. This enables anticipatory behavior and lead control that increases both gain and phase in a specific frequency range to improve system performance ([35]; see a review in Refs. [50,21]). In the target-tracking task, lead control helps participants better respond to mid-frequency perturbations in target motion where timing is critical without overreacting to high-frequency noise, enhancing gain where needed while preserving stability (Figure 1c, right panels).

The control performance metrics (such as precision, response gain, and phase lag) evaluate participants' overall control performance. These measurements, however, cannot reveal how visual cues, training, and diseases affect the sensory-motor system underlying visuomotor control. Computational modeling of the human control behavior as the human operator transfer function (Y_p , Figure 1b) can help address this issue. Particularly, the Crossover model [38,36], a quasi-linear model from classical control theory, can successfully describe

frequency-domain behavior of human operators when performing a variety of closed-loop manual control tasks such as driving and piloting aircraft (e.g. [37,46,52]). In this model, Y_p is given by:

$$Y_p = \frac{K_p e^{-s\tau} (sT_L + 1)}{s^2/\omega_n^2 + 2s\xi_n/\omega_n + 1} \quad (1)$$

The parameters like gain (K_p), reaction time (τ), and lead time constant (T_L) capture the visual-stimulus-dependent characteristics of visuomotor control and indicate control responsiveness, delay, and sensory predictions of the current control actions. A second-order low-pass filter ($\frac{1}{s^2/\omega_n^2 + 2s\xi_n/\omega_n + 1}$) captures the visual-stimulus-independent characteristics of the neuromuscular-plus-controller system, which includes a damping ratio ξ_n and an undamped natural frequency ω_n , and s is the Laplace transform variable. By separating the control responses into visually and non-visually driven components, this model highlights how perceptual and motor systems interact.

Compared to alternative control models (such as the Bayesian state feedback control model [51]), the Crossover model relies on simple control laws and does not require a full internal model of the system. Despite its simplicity, it offers a powerful framework for approximating human closed-loop control behavior, providing strong explanatory power with minimal computational complexity. In the following sections, I describe how applying classical control theory methods advances our understanding of visuomotor integration, revealing how cues, training, and neurological conditions shape performance, with implications for real-world applications in rehabilitation and interface design.

Using control theory to study human visuomotor control abilities

Normal adults

Control theory methods have been instrumental in uncovering how the visual system processes information to guide motor control, shedding light on the neural mechanisms linking perception and action. A long-standing question in this domain is how variations in visual stimuli influence motor performance, especially given the proposed separation of visual pathways for perception and action (e.g. [17,39]).

To address this question, Li et al. [31,32] examined whether visual cues such as luminance and color contrast, known to affect passive perception, also influence active control. Using target-tracking tasks, they found that manual control of a moving stimulus depends as much on these visual cues as passive perception, suggesting a shared early-stage visual processing system that supports both perceptual and motor functions. In a

Table 1**Summary of relevant concepts of control theory methods.**

Term	Definition
Bayesian inverse optimal control analysis	A computational framework that uses Bayesian inference to estimate the underlying objectives driving human behavior, assuming near-optimal performance. By analyzing observed actions, it infers latent control strategies, such as energy efficiency or stability, offering insights into decision-making processes in motor and cognitive tasks.
Closed-loop feedback	A control system where the system's output is fed back into the input to regulate and optimize the system's performance. The feedback signal may be the output signal itself or a function of the output signal and its derivatives and/or integrals.
Compensatory control models	Models focus on how systems adjust their behavior to maintain stability in the face of disturbances or changes, such as humans adjusting motor responses to compensate for visual disruptions.
Damping ratio	A parameter that describes how oscillations in a system decay over time. For example, a car suspension system with a good damping ratio will stop bouncing quickly after hitting a bump.
Differentiator	The output signal of a differentiator is the first derivative of the input signal, generally multiplied with a constant factor. If the input waveform is non-changing, the output of the differentiator is zero. An abrupt signal change leads to a large output of the differentiator.
Frequency-based analysis	A method of analyzing how systems respond to input signals of different frequencies. In control theory, it helps determine how well a system handles slow or fast changes.
Gain controller	Gain controller establishes the proportional relationship between the magnitude of input (error) and controller output. The controller adjusts the gain (proportional coefficient) to amplify or attenuate the input signal, thereby influencing the system's response.
Human-in-the-loop systems	Human-in-the-loop systems involve continuous interaction between humans and automated systems, where human input (such as sensory data and motor actions) influences the system's performance. These systems are studied in areas like driving simulators and sports training, with a focus on optimizing performance through real-time feedback and control mechanisms.
Laplace transform variable	The Laplace transform is a mathematical tool used to analyze dynamic systems. The variable, often represented as "s," helps convert time-based equations into frequency-based equations for easier analysis.
Lead control	A control strategy that is technically achieved by a phase lead of input signals using derivatives with no internal models. Lead control selectively increases both gain and phase in a specific frequency range to improve system performance.
Response gain	The magnitude increase of the system output in response to the input changes.
Second-order low-pass filter	A system that attenuates high-frequency signals while allowing low-frequency components to pass through with minimal reduction. It is characterized by a quadratic transfer function that defines how the filter responds to different frequencies, often including parameters such as natural frequency and damping ratio to control its behavior. Compared to first-order filters, a second-order filter provides a steeper roll-off, meaning it more effectively reduces unwanted high-frequency noise while maintaining the integrity of lower-frequency signals.

related study, W. O. Li et al. [29] examined how the introduction of new sensory cues affects control performance. They found that novel visual cues can enhance performance only when they complement the existing sensory feedback cues that are appropriate for the type of plant control dynamics (such as position cues for velocity control and motion cues for acceleration control). These findings have practical implications for designing feedback systems, such as driver assistance technologies, where additional visual cues must be carefully integrated to support rather than disrupt existing plant control dynamics.

In another application, Li et al. [26] investigated the potential of visuomotor coordination training, such as playing action video games, to enhance visuomotor control. They found that after only 5 to 10 hours of action video game training, nonaction gamers demonstrated significant improvements in performance on both lane-keeping and target-tracking tasks. Modeling performance with the Crossover model revealed

that even brief exposure (five hours) to action video games enhanced the sensorimotor system's responsiveness to input error signals. First-person-shooter games, in particular, also improved the system's ability to anticipate errors and generate lead control. This suggests that action gaming enhances the efficiency of sensorimotor integration, making it a promising tool for training visuomotor skills in real-world tasks such as driving.

Control theory has also advanced our understanding of the control of self-motion tasks, such as lane keeping. Lane-keeping control involves continuously adjusting steering to minimize lateral deviation, a process that can be studied through simulated crosswind perturbations [18,19,3]. Li and J. Chen [27] investigated the contributions of various visual cues (such as optic flow, splay angle, and bearing angle) to lane keeping by simulating driving scenarios with varying types of visual information. Their control systems analysis revealed that dense flow information (i.e. the pattern of apparent motion of

objects, surfaces, and edges in a visual scene caused by the relative motion between an observer and the scene) improved control precision and response gain, but when splay angles were available, participants prioritized them over other cues. These results emphasize the importance of integrating multiple visual references, such as road markings and roadside objects, in highway and simulator designs to enhance driver safety.

Another fundamental self-motion task is goal-oriented locomotion, where individuals must walk or steer toward a target based on available visual information. Although previous studies have shown that people rely on both optic flow and target visual direction relative to body orientation to perform this task [43,49], how these two cues interact with each other during goal-oriented locomotion remains largely unknown. Li and Niehorster [28] used a control theory approach and, for the first time, separated the continuous steering control response to these two cues by applying two sets of harmonically independent sum of sines to perturb these two cues, respectively (see also Ref. [47]). They found that while adding flow information improved control precision and reduced phase lag for the control response specific to optic flow, the control response specific to target visual direction remained unaffected. These findings suggest that the visual system uses these two cues independently for goal-oriented locomotion control.

Further applications of control theory include examining how optic flow cues influence the control of self-motion in dynamic environments. Li et al. [30] applied frequency-based analysis to quantify the effects of optic flow cues beyond the instantaneous velocity field (such as flow lines and acceleration) on the control of self-motion. Their findings revealed that incorporating features like flow lines and acceleration significantly improved performance at higher input frequencies. This underscores the importance of designing visual interfaces with enriched optic-flow information to enable more precise and anticipatory control.

In summary, research studies discussed in this section demonstrate the value of control theory in systematically examining how visual cues support motor control. The insights gained have practical applications in training, interface design, and real-world locomotion tasks, improving safety and performance in activities requiring complex visuomotor integration.

Athletes

For recent studies that examined visuomotor abilities in athletes, some examined athletes' visuomotor strategies for collision avoidance using virtual reality, showing that effective visual scanning strategies are critical for navigating dynamic environments [23]. Some proposed vision training, a process using visual exercises as part of a

structured sports conditioning program, for numerous sports with goals to enhance performance and reduce injury risks, particularly concussions (for a review, see [10]).

A significant application of control theory in sports science involves analyzing visuomotor control in baseball players. Using control theory, Marken [34] showed that to identify the informational basis of fly ball catching in outfielders, one must examine how optical variables respond to disturbances, revealing that behavior reflects control of perception, not reaction to cues. Hitting a baseball requires precise hand-eye coordination, yet the relationship between fundamental visuomotor abilities and batting performance remains largely unexplored. R. Chen et al. [9] investigated this link using control theory methods to assess players' basic eye tracking and manual control abilities. Their findings suggest that while baseball players possess superior fundamental visuomotor skills compared to nonathletes, these abilities primarily stem from innate traits rather than accumulated experience. However, with increased experience, the correlation between innate visuomotor skills and batting accuracy strengthens, indicating that training enhances the integration of these fundamental abilities into complex motor tasks.

R. Chen et al. developed a linear additive noise cascade model illustrating how noises from different stages of the visuomotor system contribute to the above findings. The noises include visual noise from dynamic visual motion processing (η_v), manual motor noise from manual control (η_m), oculomotor noise from eye tracking (η_o), and batting-specific noise from batting performance (η_b). Visual noise (η_v) is shared between the visual system and all three output systems (manual control, eye tracking, and batting) and sets an upper limit (i.e. rate-limiting) for all downstream performance. Batting noise (η_b) is a mutable source of variability that decreases linearly with years of baseball experience, which explains why correlations between fundamental visuomotor skills (such as eye tracking and manual control abilities) with batting performance strengthen over time (Figure 2).

These insights highlight the effectiveness of control theory approaches in providing a valuable framework for measuring and predicting athletic potential. Understanding the mechanisms behind athletes' exceptional visuomotor skills can inform tailored training strategies to improve accuracy, reaction time, and injury prevention.

Patients with neuromotor disorders

Control theory methods have played an important role in understanding visuomotor control deficits in patients with neuromotor disorders such as PD. Research suggests that these deficits primarily arise from impairments in sensorimotor processing rather than from motor

Figure 2

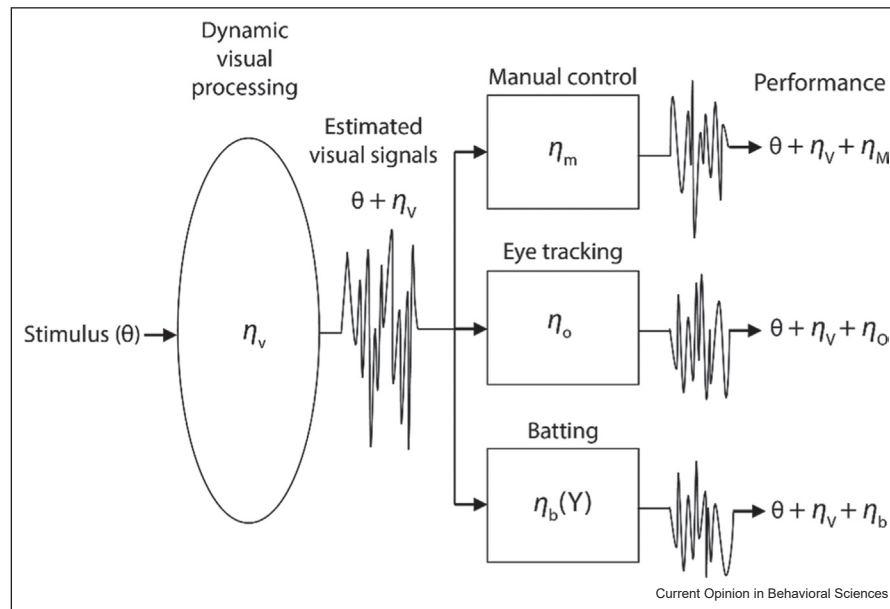


Illustration of the linear additive noise cascade model with shared rate-limiting noise in dynamic visual motion processing (η_v) and unshared noise unique to a specific output system, such as manual motor noise (η_m) that only affects manual control, oculomotor noise (η_o) that only affects eye tracking, and batting-specific noise (η_b) that only affects baseball batting. Batting noise (η_b) is a mutable source of variability that decreases linearly with years of baseball experience, which explains why correlations between fundamental visuomotor skills (such as eye tracking and manual control abilities) with batting performance strengthen over time.

Adapted from Ref. [9].

execution alone (e.g. [16,20]). Recent studies propose that PD disrupts an integrated mechanism of action regulation, affecting both movement initiation and inhibition (e.g. [53]). This highlights the need for assessments that go beyond standard clinical motor evaluations to better capture the complexities of visuomotor dysfunction in PD.

J. Chen et al. [8] applied control theory to investigate how PD and antiparkinsonian medication influence visuomotor control using the target-tracking task as depicted in Figure 1a. They found that while medication improved visuomotor control, PD patients still exhibited reduced control precision and response gain, alongside increased phase lag compared to healthy controls. Modeling performance with the Crossover model revealed that PD impaired both the responsiveness and predictive capabilities of the sensorimotor system, as well as the stability of the neuromuscular system. Medication enhanced sensorimotor responsiveness and prediction but did not restore neuromuscular stability to healthy levels.

These findings provide the first quantitative evidence of how PD and its treatment impact visual-stimulus-dependent and -independent aspects of visuomotor control. They underscore the potential of control theory in

developing sensitive assessment tools for evaluating treatment efficacy and monitoring fitness for daily tasks that require visuomotor skills, such as driving in PD patients. This echoes the recent applications in systems biology, showing that control theory approaches offer a promising framework for understanding disease mechanisms and identifying therapeutic targets [1].

Control theory and continuous psychophysics

In recent years, continuous psychophysics — a research approach in which the stimulus varies continuously over time and participants provide continuous responses to the stimulus — has emerged as a valuable approach to studying human visual perception. Unlike traditional trial-based psychophysics that focuses on discrete stimuli and responses, continuous psychophysics captures real-time interactions between perception and behavioral responses such as eye movements and manual tracking [24,6] and provides insights into the temporal dynamics of visual processing [7]. Researchers have successfully used continuous psychophysics to map participants' behavioral responses to object and self-motion stimuli [14,22]. When combined with Bayesian inverse optimal control analysis, researchers have also used continuous physics to estimate perceptual thresholds, action variability, and subjective beliefs about stimulus dynamics [45]. Although most developed in visual

perception research (e.g. tracking moving targets), continuous psychophysics is emerging in auditory and multisensory domains (e.g. real-time tracking of sound location or pitch in dynamic listening tasks applies the same principles).

Continuous psychophysics aligns closely with control theory in its emphasis on the ability of perceptual systems to rapidly and accurately respond to continuously changing stimuli with minimal delay and error. This alignment is not just conceptual but reflects a shared mathematical structure. Continuous psychophysics can be concretely framed as a control problem, in which participants function as human operators who continuously estimate stimulus properties (e.g. position, velocity) and adjust behavioral responses to minimize perceptual error. This framing invites the application of control-theoretic models and tools to capture how perceptual estimates evolve over time and how responses are updated based on both current sensory input and sensory predictions. For example, optimal filtering techniques in control theory, like Kalman filters, have been used in continuous psychophysics to model visual sensitivity and explain how perceptual systems cope with noisy, time-varying inputs [14,5].

In summary, the relationship between continuous psychophysics and control theory is symbiotic. The former offers empirical insights into dynamic perceptual processes, and the latter provides robust mathematical frameworks for modeling and prediction. Merging continuous psychophysics with control theory presents new opportunities to study perception and action as an integrated, closed-loop system. Control-theoretic tools (e.g. system identification, frequency-domain analysis, closed-loop system feedback) allow precise analysis and modeling of how perception translates into action over time, capturing response gains, delays, and feedback dynamics that traditional psychophysics misses. This allows researchers to infer latent perceptual states and characterize how perception is used to guide action in real time.

Conclusions, open questions, and future directions

This review highlights recent advancements in applying control theory to the study of human visuomotor control across diverse populations, including normal adults, athletes, and patients with neuromotor disorders. Classical control theory offers a simple mathematical framework (such as the Crossover model) to account for human closed-loop behavior through feedback loops linking sensory input to motor output. It facilitates system identification from behavioral data without requiring detailed assumptions about internal representations. This approach has been shown to be effective in

accounting for dynamic behavior in continuous target-tracking tasks, revealing how visuomotor systems respond to input errors and generate commands taking control dynamics into consideration. The review synthesizes findings showing how visual cues support motor control in normal adults, how control-theoretic approaches offer a robust framework for assessing and predicting athletic potential, and how PD and its treatment affect both visual-stimulus-dependent and -independent aspects of visuomotor control. Together, these results demonstrate the explanatory power and practical relevance of classical control theory in uncovering the mechanisms underlying visuomotor performance.

While much of the control theory research has focused on target-tracking tasks with manual responses, similar principles apply to target tracking with eye movements. Niehorster et al. [40] found that concurrent manual tracking enhances smooth pursuit eye tracking even for unpredictable target motion. However, the mechanisms underlying this enhancement and how visual input is shared between eye and hand motor systems remain largely unknown. Despite the long-standing proposal to apply control systems analysis to oculomotor research [42], recent progress in using control theory for eye movement studies has been limited, presenting new opportunities for future investigations.

Human motor control can be modeled through a range of approaches (see Ref. [15]). In the context of visuomotor control, classical control theory models the brain or the observer as an agent that continuously adjusts motor output based on sensory feedback to achieve a desired goal. Its strength lies in capturing real-time, closed-loop dynamics using well-defined mathematical tools such as transfer functions and frequency response analyses.

Another control theory approach, perceptual control theory, also models human behavior as a closed-loop feedback system. However, different from classical control theory, it emphasizes the idea that humans act to control their perceptions, not just to respond to external stimuli [34]. Perceptual control theory posits that actions emerge from the dynamic interaction between an individual and the environment, driven by subjective perceptual goals. This alternative approach focuses on identifying perceptual variables and their coupling rules that give rise to control behavior [13,2,33]. Future research should explore the interplay between the processing of external visual stimuli and internal perceptual goals to better understand visuomotor control. Integrating these perspectives may enhance our ability to model empirical data, ultimately informing the development of advanced training programs and assistive technologies to improve human visuomotor control across various domains.

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Data Availability

No data were used for the research described in the article.

Declaration of Competing Interest

The author declares that there were no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Declaration of Generative AI and AI-assisted technologies in the writing process

During the preparation of this work, the author used ChatGPT to improve language and readability. After using this tool, the author reviewed and edited the content as needed and takes full responsibility for the content of the publication.

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